

-Origin story:

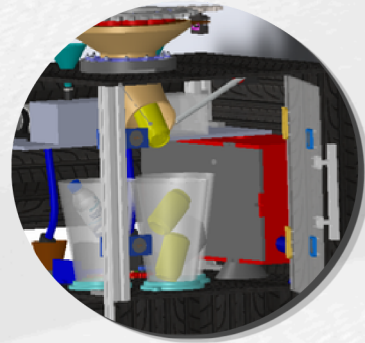
-This project started with my mother — torn every day between work and endless household chores: sorting waste, cleaning, disinfecting, with no time left for herself. MONGEYA is my answer to that: a machine that takes over the repetitive, physical work at home.



1. Automatic Waste Sorting

A camera above the hopper identifies each item. A motorized arm on a rail then guides it onto a rotating platform with three compartments.

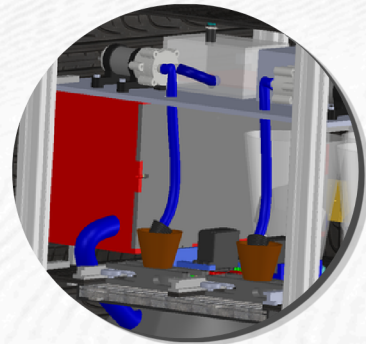
- Camera-based detection
- Motorized sorting arm on rail
- Rotating platform: aluminum / plastic / paper



2. Automatic Chemical Refill

Two independent pumps draw from internal tanks and dispense bleach and cleaning gel with precise dosing — no manual handling, no overdose risk.

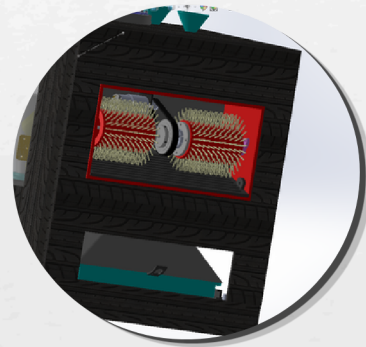
- 2 independent dosing pumps
- Dedicated bleach / cleaning gel tanks
- Calibrated internal tubing



3. Shoe Cleaning Station

A rotating brush scrubs the shoe while a built-in vacuum captures dust and debris in real time.

- Rotating brush
- Simultaneous vacuum extraction
- Front-access compartment

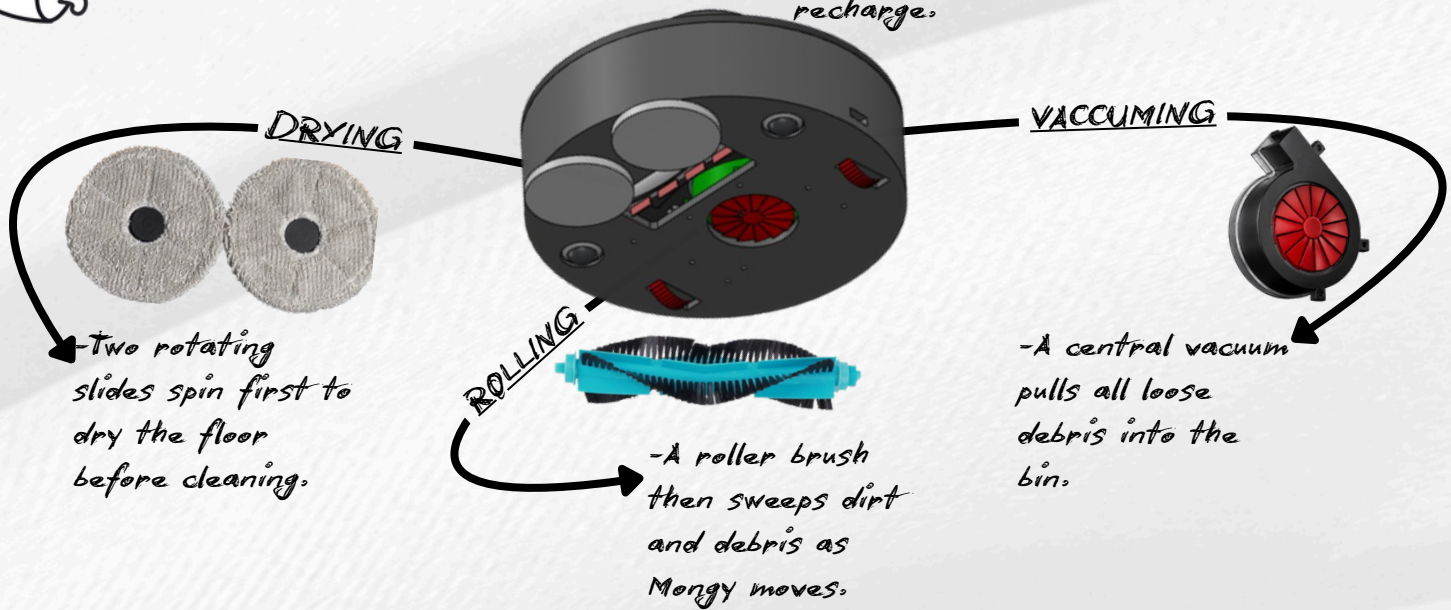


MONGY

HACK CLUB

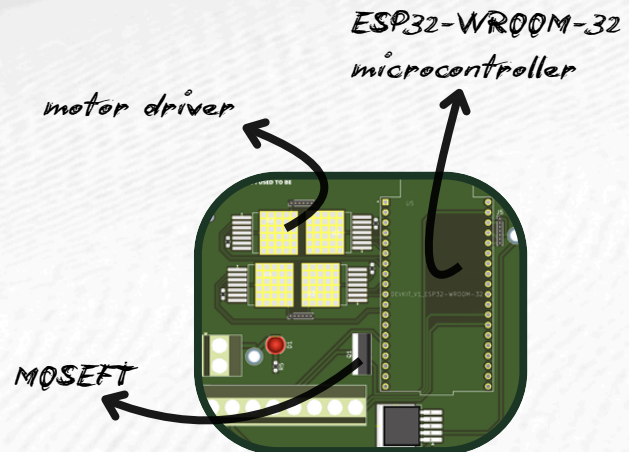
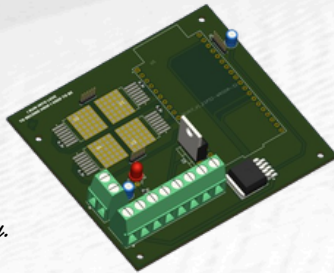
-The son of MONGEYA — leaves the "belly" to clean the floor

-When MONGEYA detects debris nearby, Mongy rolls out, cleans, dries, then returns to dock and recharge.

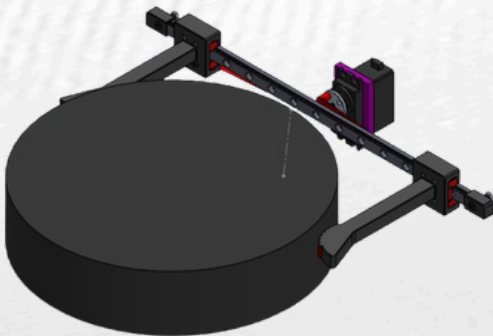


-CUSTOM PCB

A custom board (KiCad) built around an ESP32-WROOM-32 optimizes the limited space inside Mongy.



-DOCKING & PROTECTION



-A mechanical locking system holds Mongy firmly inside MONGEYA during transport, charging, and standby.

-NAVIGATION



-LIDAR for real-time mapping and obstacle avoidance
-PID control with motor encoders for precise trajectory